Exoplaneten

1

Gliederung

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3. Kontakt mit extraterrestisches Leben
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5. Habitable Zone
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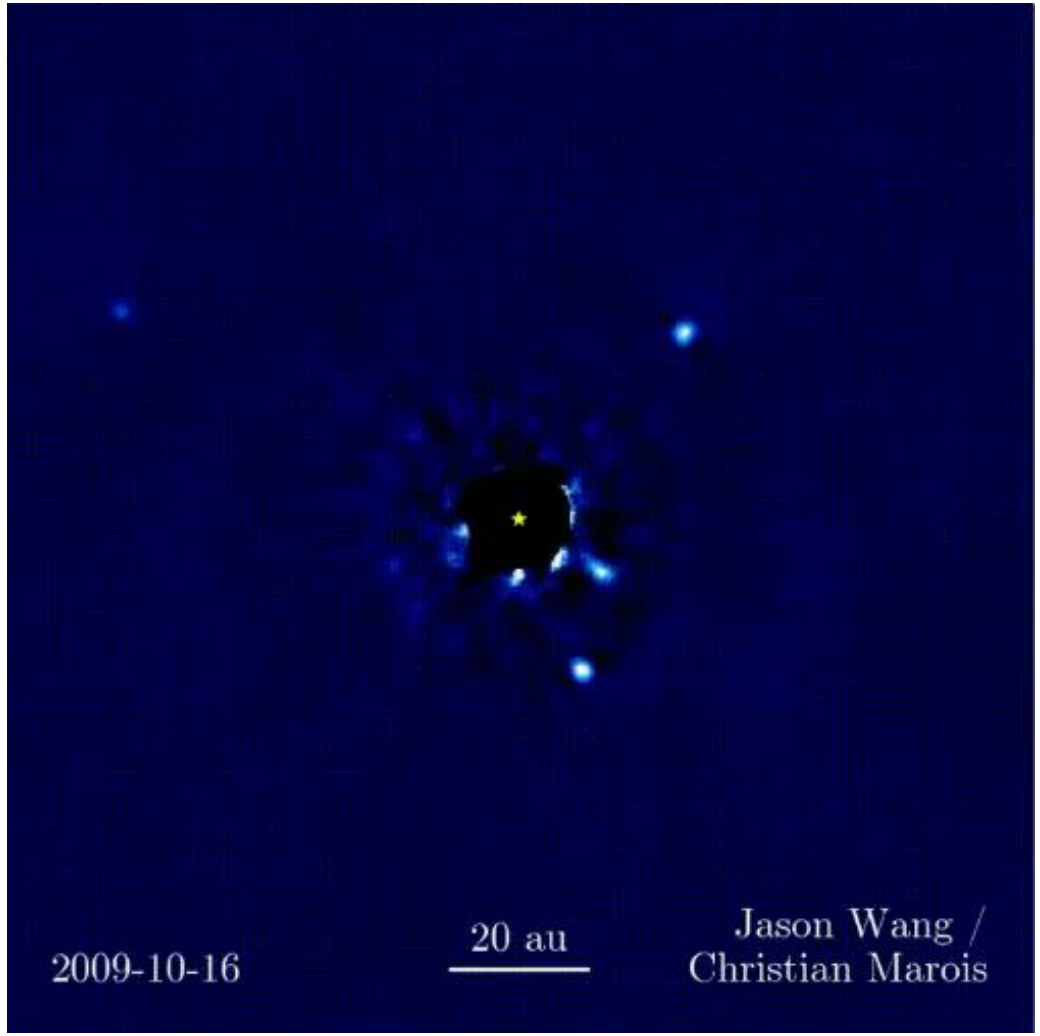
Direkte Beobachtung

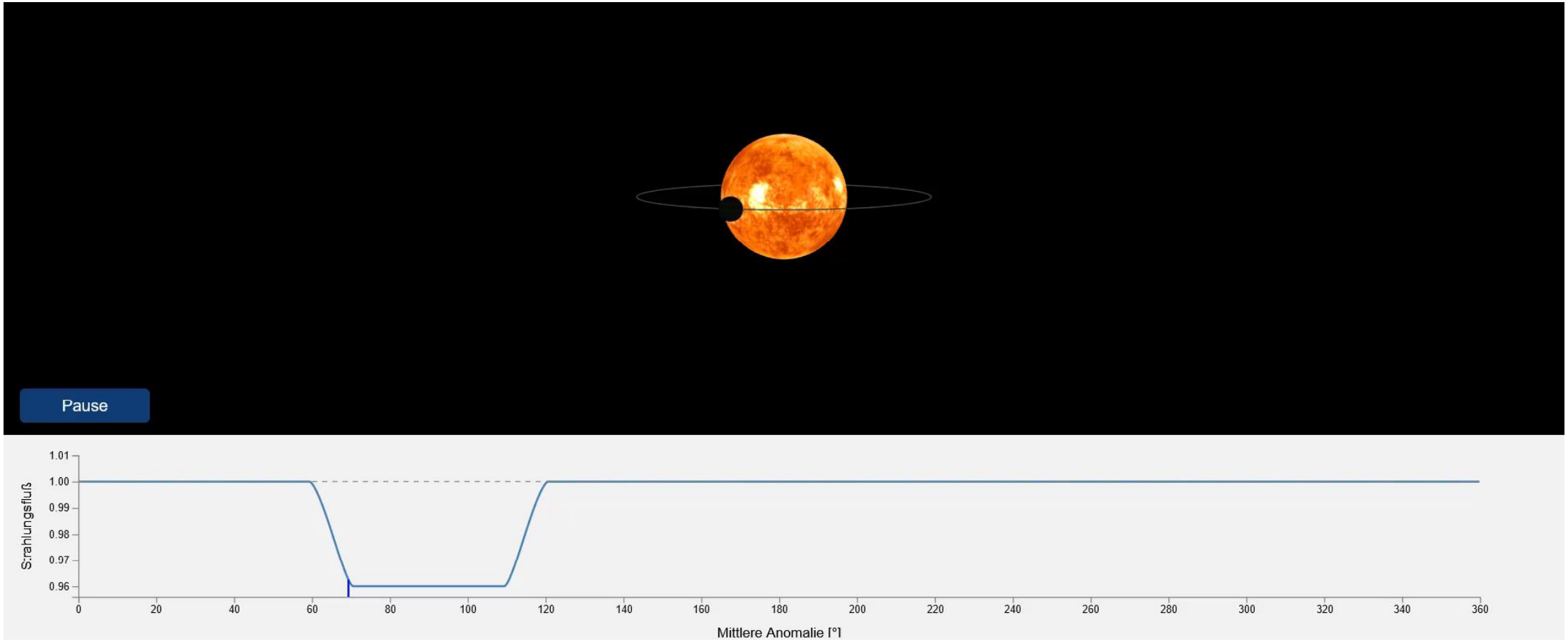
Planetensystem HR 8799:



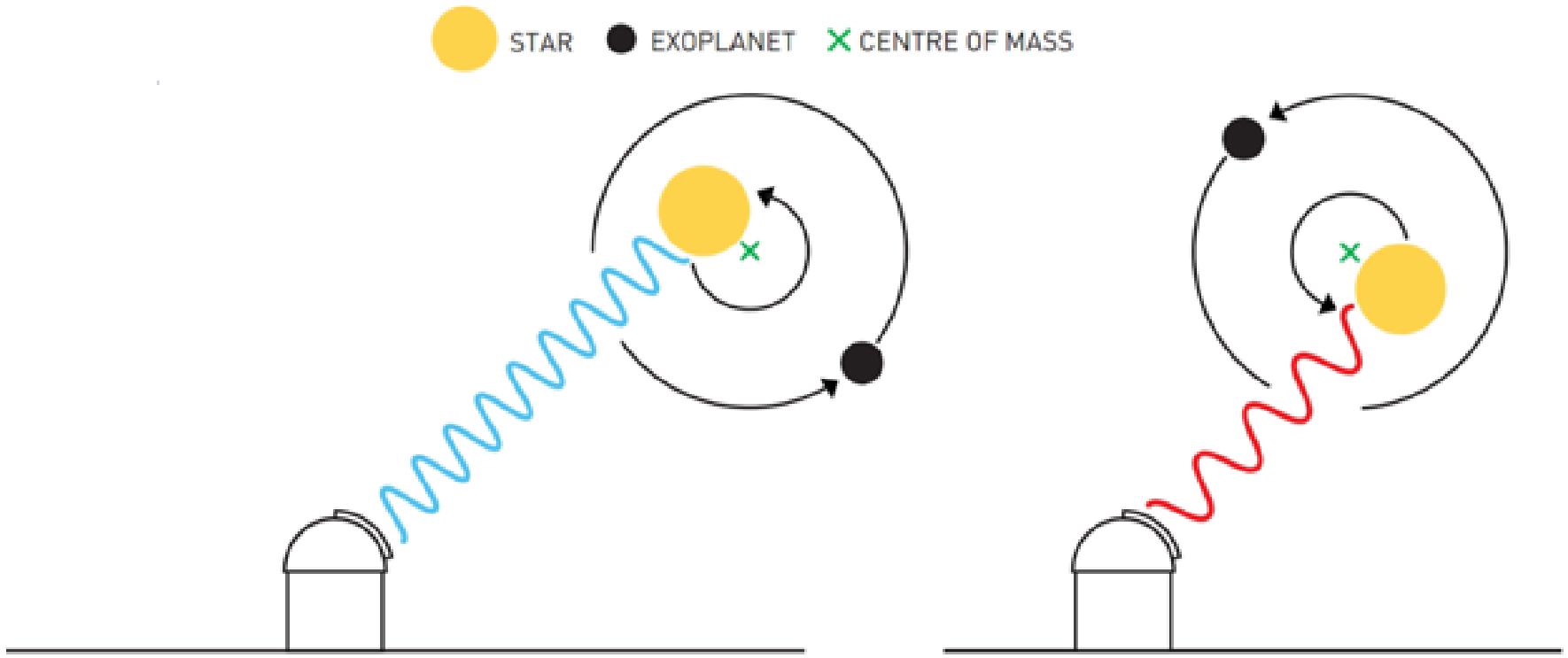
Indirekte Nachweismethoden

- Erfolgreichste:
  - Transitmethode
  - Radialgeschwindigkeitsmethode
- Weitere:
  - Astrometrische Methode
  - Gravitational-microlensing-Methode
  - Berechnung nach gestörter Planetenbahn
  - Lichtlaufzeit-Methode

Transitmethode



Radialgeschwindigkeitsmethode



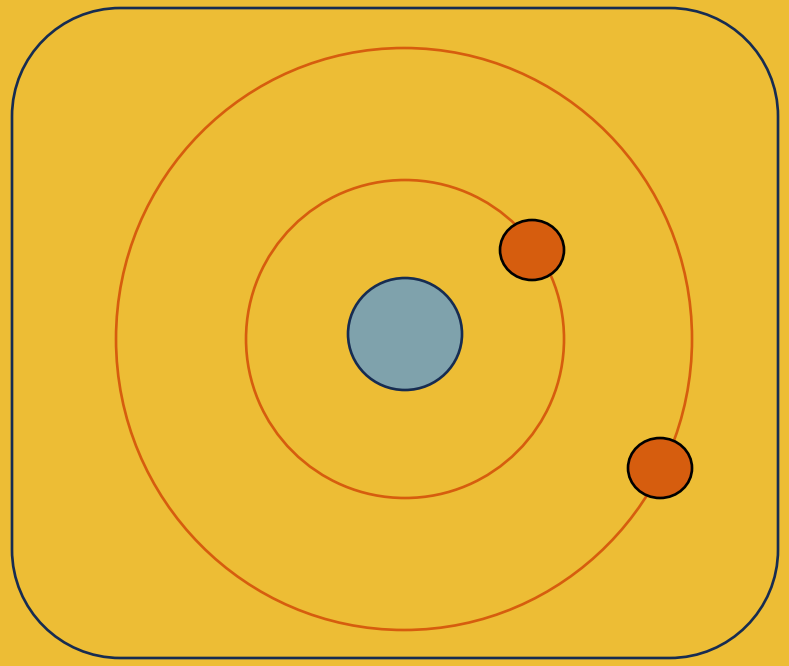
Kontakt mit extraterrestrischem Leben

Klima

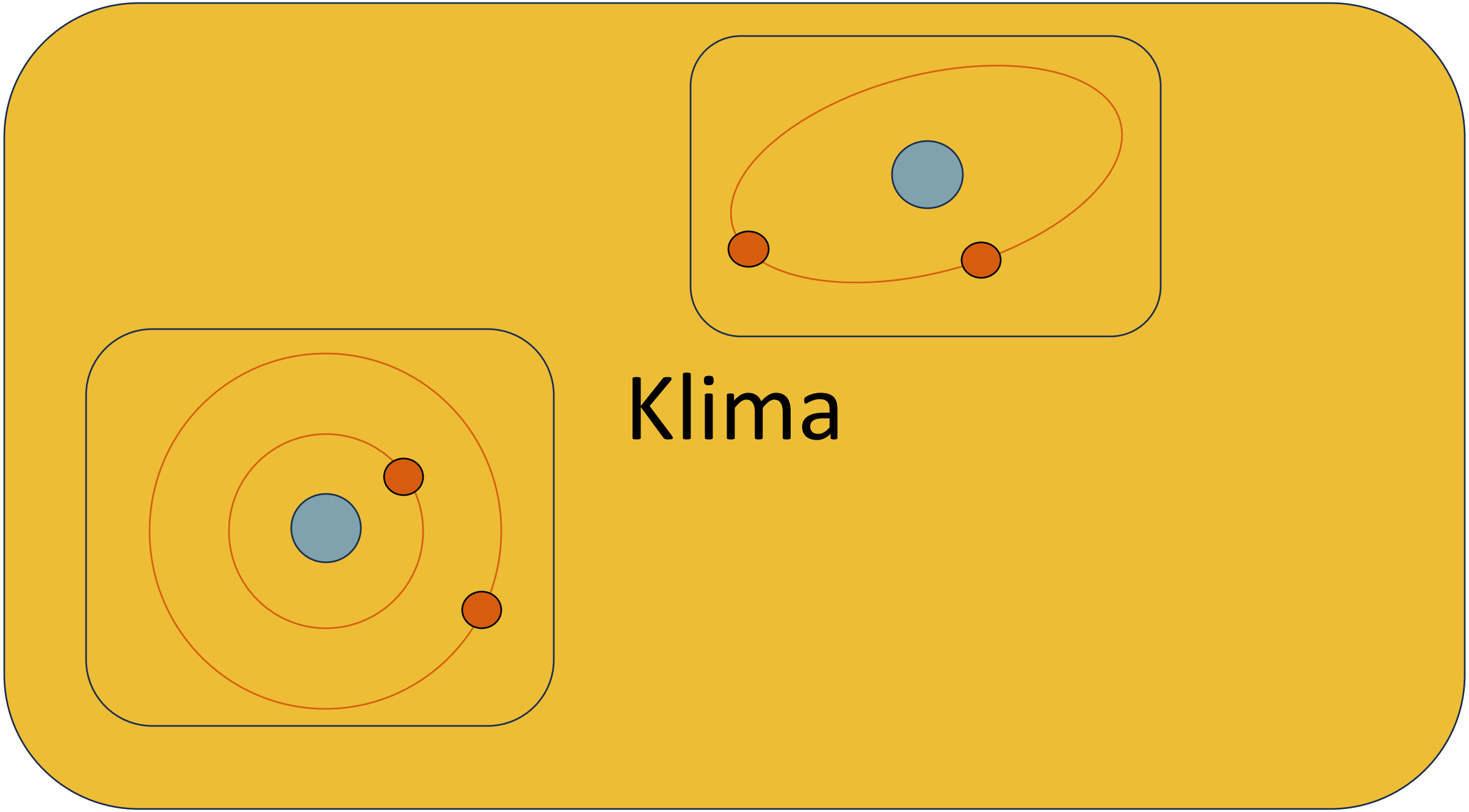
Wasser

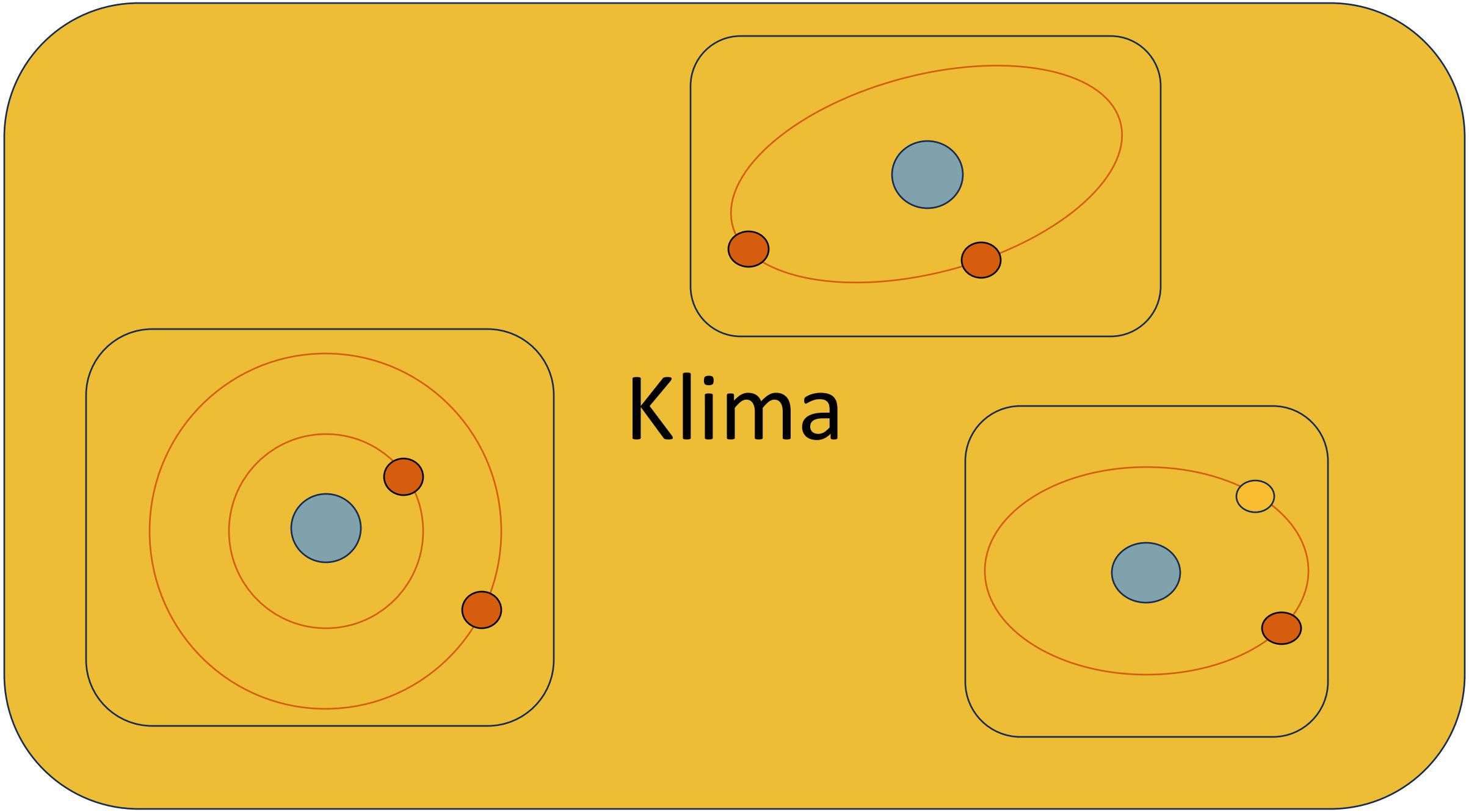
Gase

Klima



Klima


Klima


Klima

Wasser

Gase

Notwendig

H-S-H

H-O-H

Gase

Gase

Treibhausgase

O=C=O

H-O-H

Notwendig

H-S-H

H-O-H

Gase

Treibhausgase

O=C=O

H-O-H

Notwendig

H-S-H

H-O-H

Einfaches Leben

H

H-C-H

H

H-H

O=C=O

N=N

Gase

Treibhausgase

O=C=O
H-O-H

Notwendig

H-S-H
H-O-H

Einfaches Leben

H
H-C-H
H
H-H
O=C=O
N=N

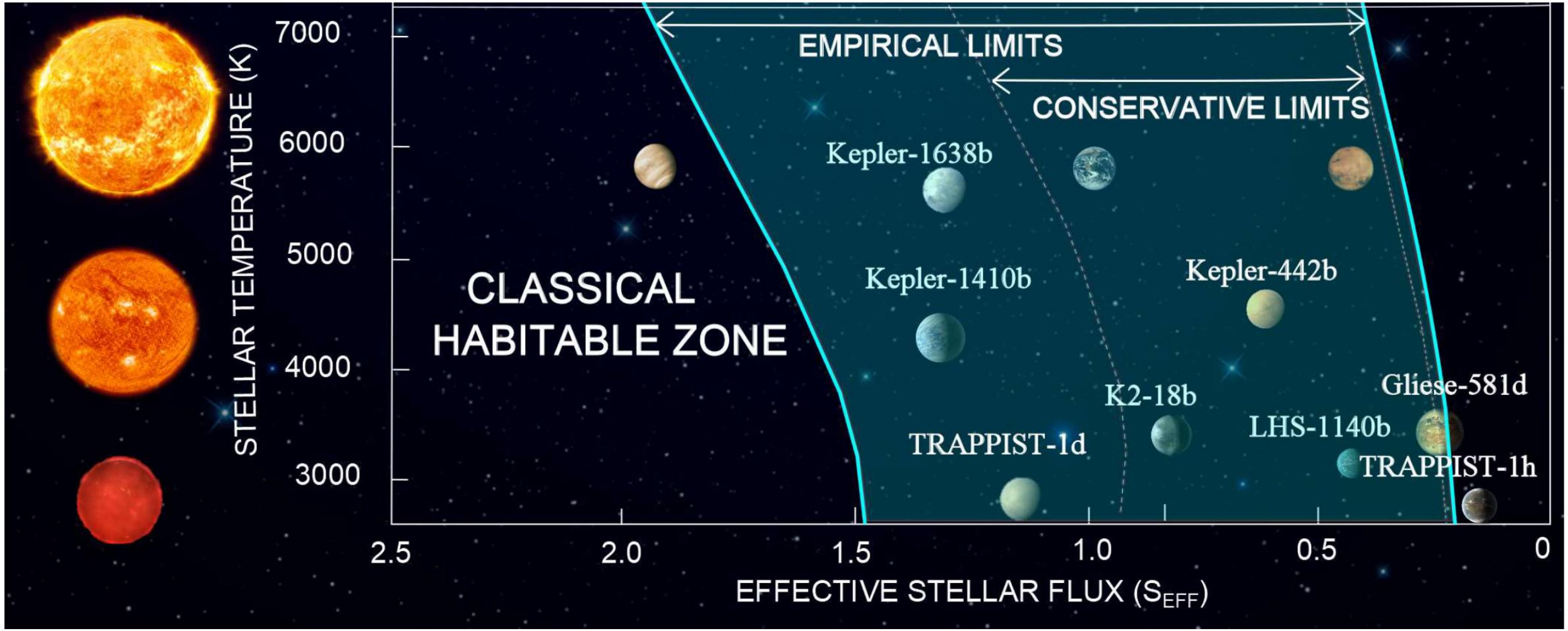
Komplexes Leben

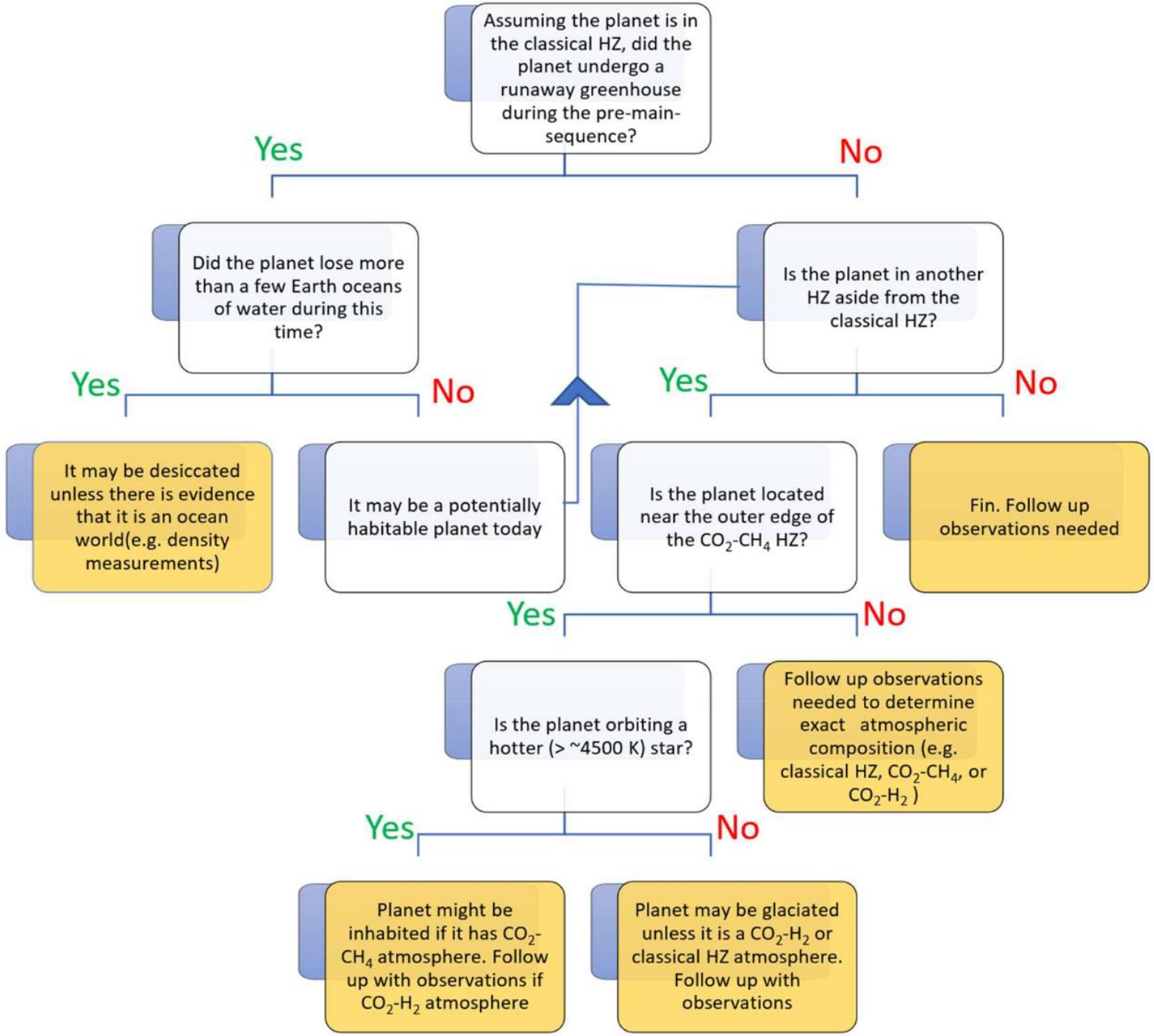
O=O
N=N

Die Habitable Zone

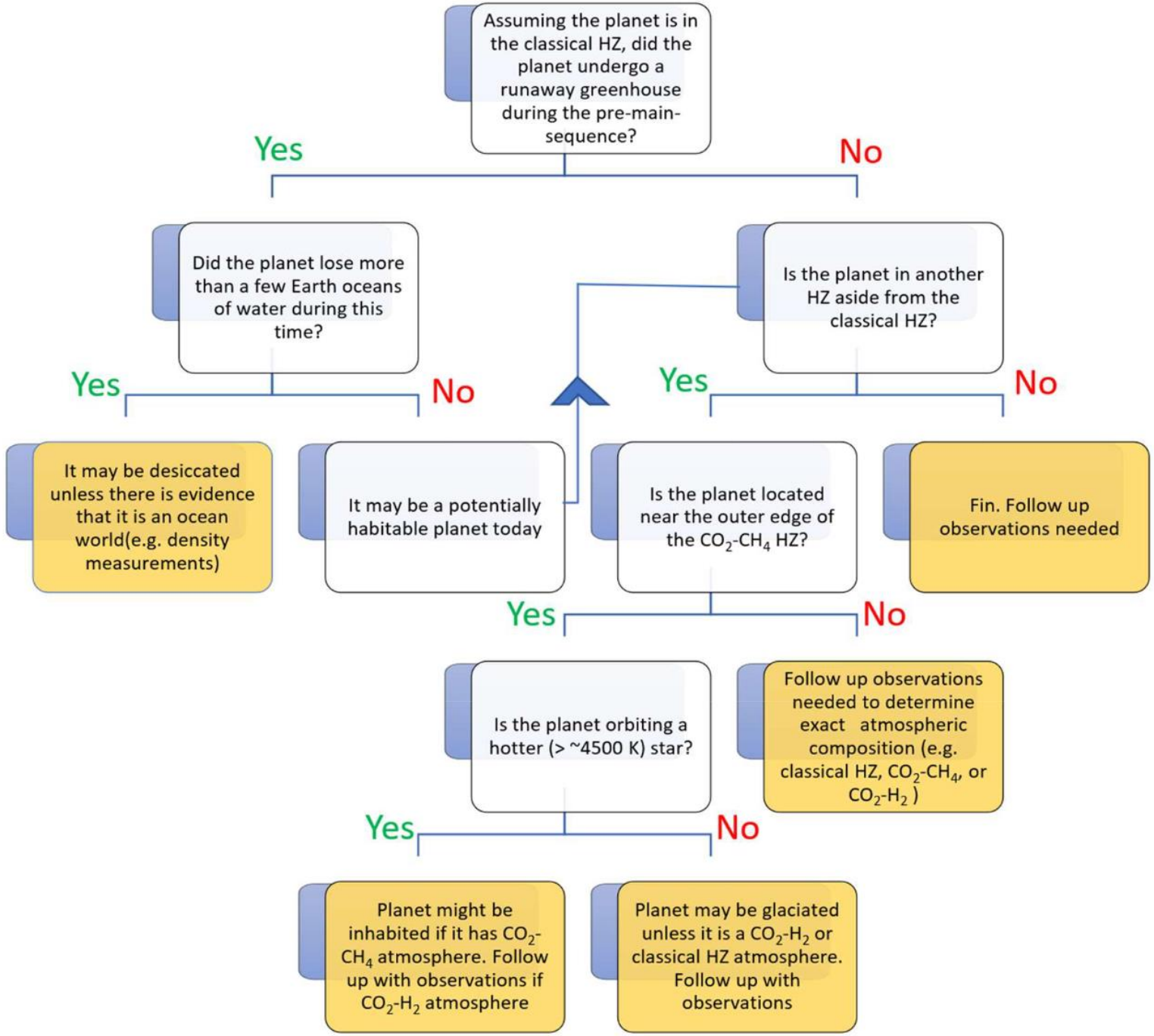
Die Habitable Zone

Region um einen Stern, in welcher Wasser in fluessiger Form existieren kann


EMPIRICAL LIMITS
CONSERVATIVE LIMITS
CLASSICAL HABITABLE ZONE
Kepler-1638b
Kepler-1410b
Kepler-442b
K2-18b
Gliese-581d
LHS-1140b
TRAPPIST-1d
TRAPPIST-1h
STELLAR TEMPERATURE (K)
7000
6000
5000
4000
3000
2.5
2.0
1.5
1.0
0.5
0
EFFECTIVE STELLAR FLUX (S_{EFF})


Assuming the planet is in the classical HZ, did the planet undergo a runaway greenhouse during the pre-main-sequence?
Yes
No
Did the planet lose more than a few Earth oceans of water during this time?
Is the planet in another HZ aside from the classical HZ?
Yes
No
Yes
No
It may be desiccated unless there is evidence that it is an ocean world(e.g. density measurements)
It may be a potentially habitable planet today
Is the planet located near the outer edge of the CO_2-CH_4 HZ?
Fin. Follow up observations needed
Yes
No
Is the planet orbiting a hotter (> ~4500 K) star?
Follow up observations needed to determine exact atmospheric composition (e.g. classical HZ, CO_2-CH_4, or CO_2-H_2)
Yes
No
Planet might be inhabited if it has CO_2-CH_4 atmosphere. Follow up with observations if CO_2-H_2 atmosphere
Planet may be glaciated unless it is a CO_2-H_2 or classical HZ atmosphere. Follow up with observations

Reisen


Assuming the planet is in the classical HZ, did the planet undergo a runaway greenhouse during the pre-main-sequence?
Yes
No
Did the planet lose more than a few Earth oceans of water during this time?
Is the planet in another HZ aside from the classical HZ?
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No
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Quellen

- https://en.wikipedia.org/wiki/Exoplanet - 24/12/13 - 8:00
- https://www.youtube.com/watch?v=EUU0-ZpFoK4 - 24/12/13 - 8:00
- https://science.nasa.gov/gallery/exoplanets/ - 24/12/13 - 8:25
- R. M. Ramirez, “A more comprehensive habitable zone for finding life on other planets,” *Geosciences*, 28 July 2018. https://www.mdpi.com/2076-3263/8/8/280 (24/12/13-8:35)
- https://en.wikipedia.org/wiki/Habitable_zone - 24/12/22 - 13-40
- https://phl.upr.edu/hwc - 24/12/22 - 13:50

Bildquellen

- https://beltoforion.de/de/exoplaneten/images/HR_8799_Orbiting_Exoplanets.gif (Folie 3, Abb. 1, 29.12.2024)
- https://beltoforion.de/de/exoplaneten/images/tyc-8998-760-1.webp (Folie 3, Abb. 2, 29.12.2024)
- https://astro-apps.org/TransitMethod/index.html (Folie 5 Abb., 29.12.2024)
- https://www.haus-der-astronomie.de/4145602/03-exoplaneten-methoden-handreichung.pdf (Folie 6, Abb., 29.12.2024)
- https://external-content.duckduckgo.com/iu/?u=https%3A%2F%2Fastrobites.org%2Fwp-content%2Fuploads%2F2021%2F03%2FScreen-Shot-2021-03-19-at-8.38.30-AM.png&f=1&nofb=1&ipt=b9ddd7d7149aebe10f6a1eae2127f4b5014aa8b2cb2e9c1116e49e9d8792a3f1&ipo=images